

BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

Name:

ID:

Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `~/BEE 4750 /HW 5/hw5-parker-regina-1`
Installed PlotUtils          v1.4.4
Installed GR_jll             v0.73.18+0
Installed OpenBLAS32_jll     v0.3.29+0
```

```

Installed Measures          v0.3.3
Installed OpenSSL           v1.6.0
Installed StaticArrays      v1.9.15
Installed MutableArithmetics v1.6.7
Installed HiGHS_jll         v1.12.0+0
Installed Pango_jll         v1.57.0+0
Installed JSON              v1.3.0
Installed FFMPEG            v0.4.5
Installed StaticArraysCore  v1.4.4
Installed GR                v0.73.18
Installed DataStructures    v0.19.3
Installed METIS_jll         v5.1.3+0
Installed Graphs            v1.13.1
Installed GraphRecipes      v0.5.15
Installed StatsBase         v0.34.8
Installed ChainRulesCore    v1.26.0
Installed SimpleTraits      v0.9.5
Installed HiGHS             v1.20.1
Installed JuMP              v1.29.3
Installed Adapt            v4.4.0
Installed Interpolations    v0.16.2
Installed StableRNGs        v1.0.4
Installed FFMPEG_jll        v8.0.0+0
Installed StructUtils       v2.6.0
Installed ForwardDiff       v1.3.0
Precompiling project...
 584.6 ms Measures
 456.2 ms ArgCheck
 465.0 ms StaticArraysCore
 469.3 ms StableRNGs
 483.5 ms Adapt
 712.2 ms DisplayAs
 660.7 ms StructUtils
 683.9 ms METIS_jll
 754.1 ms SimpleTraits
 523.0 ms EarCut_jll
1149.8 ms ChainRulesCore
1798.0 ms DataStructures
 861.7 ms OpenBLAS32_jll
1100.8 ms ColorVectorSpace → SpecialFunctionsExt
1996.3 ms OpenSSL
 636.2 ms DiffResults
2304.8 ms Pango_jll

```

752.1 ms	Adapt → AdaptSparseArraysExt
489.2 ms	OffsetArrays → OffsetArraysAdaptExt
3238.8 ms	FFMPEG_jll
726.1 ms	StructUtils → StructUtilsTablesExt
1359.4 ms	MarkdownTables
4305.1 ms	ColorSchemes
5590.2 ms	MutableArithmetics
860.8 ms	ChainRulesCore → ChainRulesCoreSparseArraysExt
1182.2 ms	LogExpFunctions → LogExpFunctionsChainRulesCoreExt
795.8 ms	SortingAlgorithms
1474.6 ms	SpecialFunctions → SpecialFunctionsChainRulesCoreExt
3139.7 ms	JSON
6338.3 ms	StaticArrays
4161.0 ms	HiGHS_jll
3789.3 ms	ForwardDiff
5616.6 ms	FFMPEG
2068.3 ms	BenchmarkTools
2845.3 ms	StatsBase
981.1 ms	StaticArrays → StaticArraysChainRulesCoreExt
672.5 ms	StaticArrays → StaticArraysStatisticsExt
1534.8 ms	ArnoldiMethod
658.9 ms	Adapt → AdaptStaticArraysExt
6376.5 ms	GR_jll
1414.7 ms	ForwardDiff → ForwardDiffStaticArraysExt
2443.1 ms	GeometryTypes
11527.1 ms	PlotUtils
4211.3 ms	Interpolations
6577.7 ms	Graphs
1552.4 ms	Interpolations → InterpolationsForwardDiffExt
3139.0 ms	PlotThemes
17631.4 ms	HTTP
4226.0 ms	RecipesPipeline
13134.3 ms	GeometryBasics
3189.5 ms	GR
1464.4 ms	NetworkLayout
1719.4 ms	NetworkLayout → NetworkLayoutGraphsExt
29512.1 ms	PrettyTables
3982.4 ms	GraphRecipes
34345.8 ms	MathOptInterface
2028.5 ms	MathOptIIS
8780.8 ms	HiGHS
32467.1 ms	DataFrames
14440.2 ms	JuMP

```

2206.7 ms    Latexify → DataFramesExt
46233.3 ms   Plots
3320.1 ms    Plots → GeometryBasicsExt
63 dependencies successfully precompiled in 78 seconds. 189 already precompiled.

```

```

using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables

```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

```
ash_frac = (.15*.08) + (.40*.03) + (.05*0.05) + (.10*0.02) + (.02*0.15) + (.05*0.02) + (.18*0.02)
recycle_frac = (.15*0) + (.40*.55) + (.05*0.15) + (.10*0) + (.02*0) + (.05*0.30) + (.18*0.40)

println("Recycling fraction = ", round(recycle_frac; digits=3))
println("Ash fraction = ", round(ash_frac; digits=3))
```

Recycling fraction = 0.397

Ash fraction = 0.177

Solution 1.1 Because all three cities have the same exact waste composition table, their overall recycling and ash fractions will be the same. The recycling fraction was found by finding the product of each component's % of total mass and MRF recycling rate and then summing this number for all components. The ash fraction was found the same way except using the % of combustion ash instead of the MRF recycling rate. The results yielded a 17.7% ash fraction and 39.7% recycling fraction.

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

Solution 1.2

The decision variables are:

- x_i^L : mass of MSW (Mg/day) transported from city i to the landfill (LF).
- x_i^M : mass of MSW (Mg/day) transported from city i to the materials recovery facility (MRF).
- x_i^W : mass of MSW (Mg/day) transported from city i to the waste-to-energy facility (WTE).

Binary facility-activation variables:

- $y_L = 1$ if the landfill is operating, and 0 otherwise.
- $y_M = 1$ if the MRF is operating, and 0 otherwise.
- $y_W = 1$ if the WTE is operating, and 0 otherwise.

Parameters from Problem 1.1 (used in the objective and constraints):

- r : overall recycling fraction at the MRF, $r \approx 0.397$.
- a : overall ash fraction for waste sent to WTE, $a \approx 0.177$.

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

Solution 1.3

Thus our final mixed-integer linear program is:

$$\min_{x_i^L, x_i^M, x_i^W, y_L, y_M, y_W} \left(\sum_i [1.5 d_{iL} x_i^L + 1.5 d_{iM} x_i^M + 1.5 d_{iW} x_i^W + 50 x_i^L + (7 + 40 r) x_i^M + 60 x_i^W] + 2000 y_L + 1500 y_M + 2500 y_W \right)$$

where:

- d_{iL}, d_{iM}, d_{iW} are distances (km) from city i to the landfill, MRF, and WTE.
- 1.5 is the transportation cost in dollars per Mg-km.
- 50, 7, and 60 are the tipping/processing costs in dollars per Mg at LF, MRF, and WTE.
- 40 is the additional cost in dollars per Mg of material that is recycled at the MRF.

- r is the overall recycling fraction (from Problem 1.1), so the effective MRF cost per Mg is $(7 + 40r)$.
- 2000, 1500, and 2500 are the fixed daily operating costs for LF, MRF, and WTE.

The objective function minimizes the total daily cost of handling and processing all municipal solid waste generated by the three cities. The total cost has three main components:

1. **Transportation Costs**

Waste shipped from each city i to each facility (LF, MRF, WTE) incurs a transportation cost of 1.5 dollars per Mg-km.

This produces the transport cost term

$$1.5 d_{iL} x_i^L + 1.5 d_{iM} x_i^M + 1.5 d_{iW} x_i^W$$

for each city i .

2. **Processing / Tipping Costs**

Each facility charges a per-Mg fee for incoming waste:

- Landfill: 50 dollars/Mg
- MRF: 7 dollars/Mg
- WTE: 60 dollars/Mg

The MRF also requires an extra 40 dollars for every Mg of *recycled* material.

With the overall recycling fraction r from Problem 1.1, the effective MRF cost per Mg is $7 + 40r$.

Thus, the processing cost is

$$50 x_i^L + (7 + 40r) x_i^M + 60 x_i^W.$$

3. **Fixed Facility Operating Costs**

Each disposal facility has a fixed daily operating cost:

- Landfill: 2000 dollars/day
- MRF: 1500 dollars/day
- WTE: 2500 dollars/day

These apply when the facility is used, captured through binary variables y_L , y_M , and y_W .

The fixed cost term is

$$2000 y_L + 1500 y_M + 2500 y_W.$$

Combining transportation, processing, and fixed operating costs yields the total cost to be minimized:

$$\sum_i [1.5 d_{iL} x_i^L + 1.5 d_{iM} x_i^M + 1.5 d_{iW} x_i^W + 50 x_i^L + (7 + 40r) x_i^M + 60 x_i^W] + 2000 y_L + 1500 y_M + 2500 y_W$$

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Solution 1.4

The constraints for the optimization problem are:

1. City mass-balance constraints

All waste generated in each city must be sent to one of the three facilities:

$$x_i^L + x_i^M + x_i^W = Q_i$$

for each city i , where Q_i is the MSW generated (Mg/day) in city i .

2. Facility capacity constraints

- **Landfill capacity, including WTE ash:**

$$\sum_i x_i^L + a \sum_i x_i^W \leq 200 y_L$$

- **MRF capacity:**

$$\sum_i x_i^M \leq 350 y_M$$

- **WTE capacity:**

$$\sum_i x_i^W \leq 210 y_W$$

Here a is the overall ash fraction from Problem 1.1.

3. Nonnegativity and integrality

$$x_i^L \geq 0, \quad x_i^M \geq 0, \quad x_i^W \geq 0$$

$$y_L, y_M, y_W \in \{0, 1\}$$

The parameters r and a come from the waste composition table, with $r \approx 0.397$ (recycling fraction) and $a \approx 0.177$ (ash fraction).

Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

Solution 1.5

Combining the objective function and constraints from Solutions 1.2–1.4, the final mixed-integer program is:

$$\min_{x_i^L, x_i^M, x_i^W, y_L, y_M, y_W} \left[\sum_i (1.5 d_{iL} x_i^L + 1.5 d_{iM} x_i^M + 1.5 d_{iW} x_i^W + 50 x_i^L + (7 + 40r) x_i^M + 60 x_i^W) + 2000 y_L + \right.$$

subject to:

City mass balance (all waste must be allocated):

$$x_i^L + x_i^M + x_i^W = Q_i \quad \text{for each city } i$$

Facility capacity constraints:

$$\sum_i x_i^L + a \sum_i x_i^W \leq 200 y_L$$

$$\sum_i x_i^M \leq 350 y_M$$

$$\sum_i x_i^W \leq 210 y_W$$

Nonnegativity and integrality:

$$x_i^L \geq 0, x_i^M \geq 0, x_i^W \geq 0$$

$$y_L, y_M, y_W \in \{0, 1\}$$

Here, $r \approx 0.397$ is the overall recycling fraction and $a \approx 0.177$ is the ash fraction from Problem 1.1.

Using JuMP to solve this mixed-integer program, the optimal solution is:

- City 1: $x_1^L = 80, x_1^M = 20, x_1^W = 0$
- City 2: $x_2^L = 0, x_2^M = 90, x_2^W = 0$
- City 3: $x_3^L = 120, x_3^M = 0, x_3^W = 0$

Facility use:

- $y_L = 1$ (landfill open)
- $y_M = 1$ (MRF open)
- $y_W = 0$ (WTE closed)

The corresponding optimal objective value (total daily cost) is:

$$Z^* = 23231.8 \text{ dollars/day}$$

Rounded: $Z^* \approx 23232$ dollars/day.

```
using JuMP
using HiGHS      # or GLPK, Gurobi, etc.

cities = 1:3

#daily MSW generation (Mg/day) for each city
Q = Dict{1 => 100.0,    # City 1
         2 => 90.0,     # City 2
         3 => 120.0}    # City 3

#distances (km) from each city to each facility
d_L = Dict{1 => 5.0, 2 => 15.0, 3 => 13.0}    # to Landfill
d_M = Dict{1 => 30.0, 2 => 25.0, 3 => 45.0}    # to MRF
d_W = Dict{1 => 15.0, 2 => 10.0, 3 => 20.0}    # to WTE

#transportation cost (dollars per Mg-km)
c_trans = 1.5

#tipping / processing costs (dollars per Mg of waste entering the facility)
c_L = 50.0    # landfill
c_M = 7.0     # MRF base tipping
c_W = 60.0    # WTE

#extra recycling cost at MRF: 40 dollars per Mg recycled
c_recycle = 40.0

#effective MRF cost per Mg of incoming waste:
#base tipping + recycling cost applied to the recycled fraction
c_M_eff = c_M + c_recycle * recycle_frac

#fixed daily operating costs (dollars/day)
F_L = 2000.0
F_M = 1500.0
F_W = 2500.0

#facility capacities (Mg/day of incoming waste)
cap_L = 200.0
```

```

cap_M = 350.0
cap_W = 210.0

#model

model = Model(HiGHS.Optimizer)

#normal variables: Mg/day of waste from each city to each facility
@variable(model, x_L[i in cities] >= 0) # to landfill
@variable(model, x_M[i in cities] >= 0) # to MRF
@variable(model, x_W[i in cities] >= 0) # to WTE

#binary variables: whether each facility is opened
@variable(model, y_L, Bin)
@variable(model, y_M, Bin)
@variable(model, y_W, Bin)

#objective = minimize total cost (transport + tipping + recycling + fixed)
@objective(model, Min,
    sum(
        # transport cost
        c_trans * (d_L[i] * x_L[i] + d_M[i] * x_M[i] + d_W[i] * x_W[i])
        # disposal / processing costs
        + c_L * x_L[i]
        + c_M_eff * x_M[i]
        + c_W * x_W[i]
        for i in cities
    )
    + F_L * y_L + F_M * y_M + F_W * y_W
)

#constraints

#mass balance:
@constraint(model, [i in cities],
    x_L[i] + x_M[i] + x_W[i] == Q[i]
)

#landfill capacity: receives direct landfill waste + WTE ash
#WTE ash = ash_frac * (sum of waste going to WTE)
@constraint(model,
    sum(x_L[i] for i in cities) + ash_frac * sum(x_W[i] for i in cities) <= cap_L * y_L
)

```

```

)

#MRF and WTE capacity constraints (only incoming waste)
@constraint(model,
    sum(x_M[i] for i in cities) <= cap_M * y_M
)
@constraint(model,
    sum(x_W[i] for i in cities) <= cap_W * y_W
)

#solve optimization problem

optimize!(model)

println("\nTermination status: ", termination_status(model))
println("Optimal objective value (cost per day): ",
    round(objective_value(model); digits = 3))

println("\nFlows to landfill (x_L):")
for i in cities
    println("  City ", i, ": ", round(value(x_L[i]); digits = 3), " Mg/day")
end

println("\nFlows to MRF (x_M):")
for i in cities
    println("  City ", i, ": ", round(value(x_M[i]); digits = 3), " Mg/day")
end

println("\nFlows to WTE (x_W):")
for i in cities
    println("  City ", i, ": ", round(value(x_W[i]); digits = 3), " Mg/day")
end

println("\nFacility use:")
println("  Landfill open (y_L) = ", value(y_L))
println("  MRF open      (y_M) = ", value(y_M))
println("  WTE open      (y_W) = ", value(y_W))

```

Running HiGHS 1.12.0 (git hash: 755a8e027): Copyright (c) 2025 HiGHS under MIT licence terms
MIP has 6 rows; 12 cols; 24 nonzeros; 3 integer variables (3 binary)
Coefficient ranges:
Matrix [2e-01, 4e+02]

```

Cost      [6e+01, 2e+03]
Bound     [1e+00, 1e+00]
RHS       [9e+01, 1e+02]
Presolving model
6 rows, 12 cols, 24 nonzeros  0s
6 rows, 12 cols, 24 nonzeros  0s
Presolve reductions: rows 6(-0); columns 12(-0); nonzeros 24(-0) - Not reduced

Solving MIP model with:
  6 rows
 12 cols (3 binary, 0 integer, 0 implied int., 9 continuous, 0 domain fixed)
 24 nonzeros

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristic;
    I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Randomized round
    S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y => HiGHS solution;
    Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper; z => Trivial upper



| Src | Nodes |         | B&B Tree |       | Objective Bounds |         |        | Dynamic Cuts |  |
|-----|-------|---------|----------|-------|------------------|---------|--------|--------------|--|
|     | Proc. | InQueue | Leaves   | Expl. | BestBound        | BestSol | Gap    | Cuts         |  |
| J   | 0     | 0       | 0        | 0.00% | -inf             | 24062.2 | Large  | 0            |  |
| S   | 0     | 0       | 0        | 0.00% | 1500             | 23231.8 | 93.54% | 0            |  |



66.7% inactive integer columns, restarting
Model after restart has 1 row, 2 cols (0 bin., 0 int., 0 impl., 2 cont., 0 dom.fix.), and 2 rows



|   | Nodes | Leaves | Expl.   | BestBound    | BestSol | Gap   | Cuts |
|---|-------|--------|---------|--------------|---------|-------|------|
| 0 | 0     | 0      | 0.00%   | 23231.788312 | 23231.8 | 0.00% | 0    |
| 1 | 0     | 1      | 100.00% | 23231.8      | 23231.8 | 0.00% | 0    |



Solving report
Status          Optimal
Primal bound    23231.8
Dual bound      23231.8
Gap             0% (tolerance: 0.01%)
P-D integral    0.00168757873191
Solution status feasible
                23231.8 (objective)
                0 (bound viol.)
                0 (int. viol.)
                0 (row viol.)
Timing         0.01
Max sub-MIP depth 0

```

Nodes	1
Repair LPs	0
LP iterations	2
	0 (strong br.)
	0 (separation)
	0 (heuristics)

Termination status: OPTIMAL

Optimal objective value (cost per day): 23231.8

Flows to landfill (x_L):

City 1:	80.0 Mg/day
City 2:	0.0 Mg/day
City 3:	120.0 Mg/day

Flows to MRF (x_M):

City 1:	20.0 Mg/day
City 2:	90.0 Mg/day
City 3:	-0.0 Mg/day

Flows to WTE (x_W):

City 1:	0.0 Mg/day
City 2:	0.0 Mg/day
City 3:	0.0 Mg/day

Facility use:

Landfill open	(y_L) = 1.0
MRF open	(y_M) = 1.0
WTE open	(y_W) = -0.0

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in 'data/generators.csv.' In period 1, the demand is $d_1 = 1100\text{MW}$. In period 2, the demand is projected to be $d_2 = 1200\text{MW}$, but there is a 25% probability that it is \$1500 . In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in

the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

References

List any external references consulted, including classmates.